

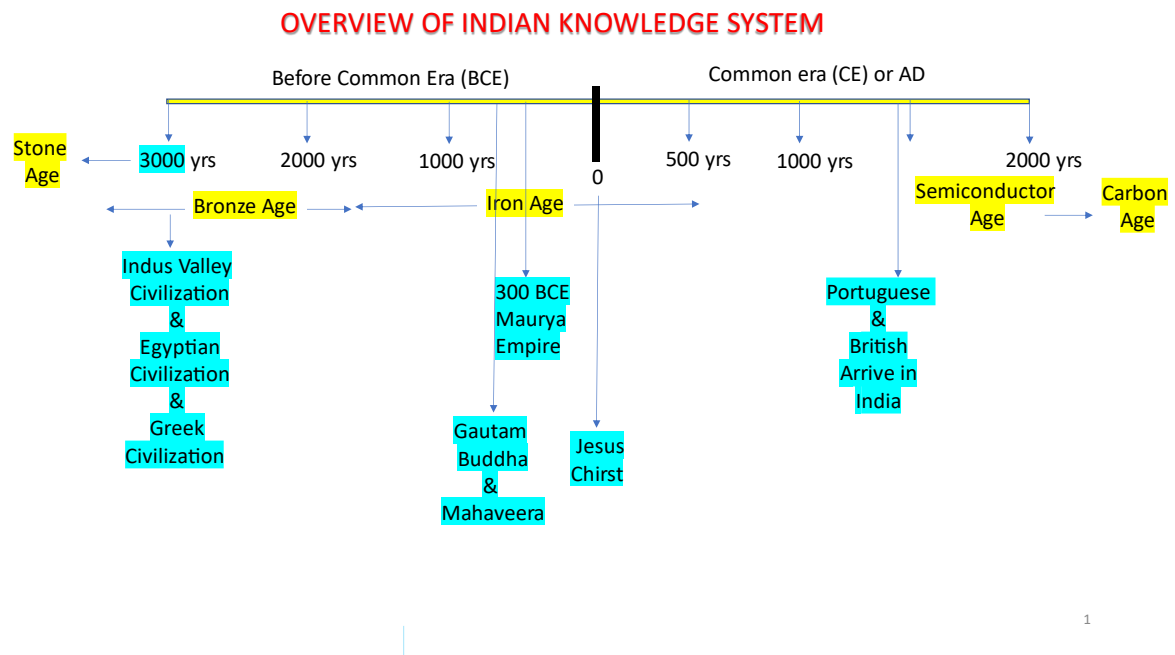


Fig.1.Indus Valley civilisation mature phase

John Marshall



Indus valley civilisation

- The history of India begins with the birth of the Indus Valley Civilization (IVC), also known as Harappan Civilization.
- It flourished around 2,500 BC, in the western part of South Asia, in contemporary Pakistan and Western India.
- The Indus Valley was home to the largest of the four ancient urban civilizations of Egypt, Mesopotamia, India and China.
- In 1920s, the Archaeological Department of India carried out excavations in the Indus valley wherein the ruins of the two old cities, viz. Mohenjodaro and Harappa were unearthed.
- In 1924, John Marshall, Director-General of the ASI, announced the discovery of a new civilisation in the Indus valley to the world.
- Three phases of IVC are:
 - the Early Harappan Phase from 3300 to 2600 BCE
 - the Mature Harappan Phase from 2600 to 1900 BCE
 - the Late Harappan Phase from 1900 to 1300 BCE.
 - The Early Harappan Phase is related to the Hakra Phase, identified in the Ghaggar-Hakra River Valley.
- The earliest examples of the Indus script date back to 3000 BC.
- This phase stands characterized by centralized authority and an increasingly urban quality of life.
- Trade networks had been established and there are also evidences of the cultivation of crops. Peas, sesame seeds, dates, cotton, etc, were grown during that time.
- By 2600 BC, the Indus Valley Civilization had entered into a mature stage.
- The early Harappan communities were turning into large urban centers, like Harappa and Mohenjodaro in Pakistan and Lothal in India.

Town Planning and Structures

- The Harappan culture was distinguished by its system of town planning.
- Harappa and Mohenjodaro each had its own citadel or acropolis, which was possibly occupied by members of the ruling class.
- Below the citadel in each city lay a lower town containing brick houses, which were inhabited by the common people.
- The remarkable thing about the arrangement of the houses in the cities is that they followed the grid system.
- Granaries constituted an important part of the Harappan cities.
- The use of burnt bricks in the Harappan cities is remarkable, because in the contemporary buildings of Egypt mainly dried bricks were used.
- The drainage system of Mohenjo-Daro was very impressive.
- In almost all cities every big or small house had its own courtyard and bathroom.
- In Kalibangan many houses had their wells.
- At sites such as Dholavira and Lothal (Gujarat), the entire settlement was fortified, and sections within the town were also separated by walls.

Agriculture

- The Harappan villages, mostly situated near the flood plains, produced sufficient foodgrains.
- Wheat, barley, rai, peas, were produced. Millets are also found from sites in Gujarat. While rice uses were relatively rare.
- The Indus people were the earliest people to produce cotton.
- While the prevalence of agriculture is indicated by finds of grain, it is more difficult to reconstruct actual agricultural practices.
- Most Harappan sites are located in semi-arid lands, where irrigation was probably required for agriculture.
- Traces of canals have been found at the Harappan site of Shortughai in Afghanistan, but not in Punjab or Sindh.

- Although the Harappans practised agriculture, animals were also reared on a large scale.

Economy

- The importance of trade in the life of the Indus people is witnessed by the presence of numerous seals, uniform script and regulated weights and measures in a wide area.
- The Harappans carried on considerable trade in stone, metal, shell, etc.
- Metal money was not used and trade was carried by barter system.
- They practised navigation on the coast of the Arabian Sea.
- They had set up a trading colony in northern Afghanistan which evidently facilitated trade with Central Asia.

Crafts

- The Harappans were very well acquainted with the manufacturing and use of Bronze.
- Copper was obtained from the Khetri copper mines of Rajasthan and Tin was possibly brought from Afghanistan.
- Textile impressions have also been found on several objects.
- Huge brick structure suggest that brick-laying was an important craft. This also attests the existence of a class of masons.
- The Harappans practised boat-making
- The goldsmiths made jewellery of silver, gold and precious stones.
- The potter's wheel was in full use, and the Harappans produced their own characteristic pottery, which was glossy and shining.

Institutions

- Very few written materials have been discovered in the Indus valley and the scholars have not been able to **decipher** the Indus script so far.
- As a result, there is difficulty in understanding the nature of the state and institutions of the Indus Valley Civilization.
- No temples have been found at any Harappan sites. Therefore the possibility of **priests ruling** Harappa can be eliminated.
- Harappa was possibly ruled by a class of **merchants**.
- If we look for a centre of power or for depictions of people in power, archaeological records provide no immediate answers.
 - Some archaeologists are of the opinion that Harappan society had no rulers, and that everybody enjoyed equal status.
 - Another theory argues that there was no single ruler, but a number of rulers representing each of the urban centers.

Decline of the Indus Valley Civilization

- The IVC declined around 1800 BCE but the actual reasons behind its demise are still debated.
- One theory claims that **Indo-European tribe** i.e. **Aryans** invaded and conquered the IVC.
 - In later cultures various elements of the IVC are found which suggest that civilization did not disappear suddenly due to an invasion.
- On the other hand, many scholars believe **natural factors** are behind the decline of the IVC.
 - The natural factors could be **geological and climatic**.
 - It is believed that the Indus Valley region experienced several **tectonic disturbances** which causes **earthquakes**. Which also changed **courses of rivers or dried them up**.

- Another natural reason might be changes in patterns of rainfall.
- There could be also dramatic shifts in the river courses, which might have brought floods to the food producing areas.
- Due to combination of these natural causes, there was a slow but inevitable collapse of IVC.

Education in Ancient India

Early education in India commenced under the supervision of a guru.

As time progressed, due to superiority complexes, the education was imparted on the basis of caste.

• Ancient Universities

1. Nalanda University -Bihar
2. Takshashila University-Pakistan
3. Vikramashila University -Bihar
4. Valabhi University -Gujrat
5. Pushpagiri University -Odisha
6. Odantapuri University -Bihar
7. Somapura University -Bangladesh
8. Ujjain University-Madhya Pradesh
9. Kanchi University -Tamil nadu
10. Telhara university-Bihar
11. Sharada peeth university-Kashmir
12. Bikrampur university-Bengal
13. Morena Golden Tringle-Madhya Pradesh
14. Kathloor sala university-Tamil Nadu
15. Jagaddala university-Bangladesh
16. Naharjuna-Andra Pradesh

1. Nalanda University

- (also known as Nalanda International University) is located in Rajgir, near Nalanda, Bihar, India.
- It has been called “One of the earliest universities in the World.
- It was founded in the 5th century BC by the Gupta Empire, known as the ancient seat of learning.
- Contribution also was given by- The King Harshwardhan, Kumar Gupta (college of fine arts), Dharmapal Brahmin scholar (tutor), Dinnaga – Founder of the school of Logic.
- It was a Buddhist center of learning from 427 to 1197 CE.
- It became the crown jewel of the development of Buddhism in India and the world.
- First Residential International university of the world.
- It accommodated over 2,000 teachers and 10,000 students from all over the world.

- **Libraries**
- The library of Nalanda, known as **DharmaGunj (Mountain of Truth)** or **Dharmagañja (Treasury of Truth)**, was the most renowned repository of Buddhist knowledge in the world at the time.
- The library had **three main buildings- Ratnasagara, Ratnodadhi and Ratnaranjaka.**
- Each building was **nine stories high** and had an enormous **collection of books** that covered various subjects ranging from **religion, literature, astrology, astronomy, medicine** and much more.
- **Curriculum Learning** was greatly encouraged as it served **dual purposes: knowledge & practice**. Courses were drawn from every field of learning, Buddhist and Hindu, sacred and secular, foreign and native.
- Students studied Science, astronomy, medicine, and logic as diligently as they applied themselves to metaphysics, philosophy, Samkhya, Yoga-shastra, the Veda, and the scriptures of Buddhism.
- They studied foreign philosophy likewise.
- Nalanda was also the most global university of its times, attracting pupils and scholars from Korea, Japan, China, Tibet, Indonesia, Persia, and Turkey.
- **Great Scholars of Nalanda University**
Chinese traveller and scholar, **Xuanzang (Hsuan-tsang, 7th century AD)**, visited Nalanda and left a vivid account of the curriculum and of the general features of the community. Famous philosophers like **Shilabhadra, Nagarajuna**, and Mahayana were among the few scholars of Nalanda.

Fall of Nalanda University

According to the records Nalanda University was **destroyed three times** by invaders, but **rebuilt only twice**.

The first destruction was caused by the **Huns** under **Mihirakula** during the reign of **Skandagupta (455–467 AD)**. But **Skanda's successors** restored the **library** and improved it with an **even bigger building**. The second destruction came in the early **7th century** by the **Gaudas**. This time, the Buddhist king **Harshavardhana (606–648 AD)** restored the university.

- and most destructive attack came when the ancient Nalanda University was destroyed by the army led by the **Turkish leader Bakhtiyar Khilji in 1193.**

2. Takshashila University

- first University Takshashila also known as "**City of Cut Stone**" flourished from **600 BC to 500 AD.**
- Takshashila was in the kingdom of **Gandhar**, in Ancient India before partition.
- But now Takshashila is in **Rawalpindi** district of the Punjab Pakistan.

- A town located in the north-western region of India (in today's Pakistan).
- 68 subjects were taught at this university and the minimum entry age, ancient texts show, was 16.
- A wide range of subjects were taught by experienced masters: Vedas, Language, Grammar, Philosophy, Medicine, Surgery, Archery, Politics, Warfare, Astronomy, Astrology, Accounts, Commerce, Futurology, Documentation, Occult, Music, Dance, etc
- Admission into this university was purely based on merit.
- The students would opt for electives and then would do in-depth study and research into their field of choice.
- It had 10,500 students including those from Babylon, Greece, Syria, and China.
- The panel of masters at the university included legendary scholars like Kautilya, Panini, Jivak
- When Alexander's armies came to the Punjab in the fourth century BC, Takshashila had already developed a reputation as an important seat of learning.
- Thus on his return Alexander took many scholars from there with him to Greece.
- Being near the north-west frontier of India, Takshashila had to face the brunt of attacks and invasions from the north and the west.
- Thus the Persians, Greeks, Parthians, Shakas and Kushanas laid their destructive marks on this institution.
- The final blow, however, came from the Huns (also the destroyers of the Roman Empire) who, A.D. c.450, razed the institution.
- When the Chinese traveller Huen T'sang (A.D. 603-64) visited Takshashila, the town had lost all its former grandeur and international character.

3. Vikramasila University

- Was established by Dharma Pala in 8th century AD, because of the decline in scholarship of Nalanda University.
- It is a Buddhist tantric centre of learning situated in the north east of Bhagalpur, Bihar.
- According to Tibetan sources five Mahaviharas Nalanda, Vikramasila, Somapura, Odantapura, Jagaddala existed then.
- These universities had some sort of network for the exchange of knowledge.
- Vikramashila was one of the largest Buddhist universities, with more than one hundred teachers and about one thousand students.
- It produced eminent scholars who were often invited by foreign countries to spread Buddhist learning, culture and religion.
- Subjects like theology, philosophy, grammar, metaphysics, logic etc. were taught here.

- The university had 400 years of glorious years when Bukhtiyar Khilji destroyed the university during 12th century AD.
- Excavation- Meticulous excavation at the site was conducted initially by B.P. Sinha of Patna University (1960–69) and subsequently by Archaeological Survey of India (1972–82).
- It has revealed a huge square monastery with a cruciform stupa in its centre, a library building and cluster of votive stupas.
- To the north of monastery a number of scattered structures including a Tibetan and a Hindu temple have been found.
- The main stupa built for the purpose of worship is a brick structure laid in mud mortar and stands in the centre of the square monastery.
- The Vikramashila site is the place for Vikramashila Mahotsav, which is held annually during the month of February.

Indian calendar system

a. Solar Calendar:

- **Definition:** A calendar based on the Earth's orbit around the sun, measuring time by the position of the sun relative to the Earth.
- **Year Length:** Approximately 365.24 days, accounting for the time it takes for Earth to complete one orbit around the sun.

b. Lunar Calendar:

- **Definition:** A calendar based on the phases of the moon, measuring time by the moon's orbit around the Earth.
- **Month Length:** Approximately 29.53 days, the time it takes for the moon to complete one cycle of phases (from new moon to new moon).

2. Month and Year Structure

a. Solar Calendar:

- **Months:** Typically consists of 12 months, each with a fixed number of days. For example, the Gregorian calendar has months of 30 or 31 days, except February, which has 28 or 29 days.
- **Year Alignment:** The year is fixed at about 365 days (366 in leap years), closely aligning with the solar year.

b. Lunar Calendar:

- **Months:** Consists of 12 lunar months, each approximately 29.5 days long. This results in a lunar year being about 354 days long (12×29.5).
- **Year Alignment:** The lunar year is shorter than the solar year by about 11 days. To keep the lunar calendar roughly aligned with the solar year, an extra month (leap month) is periodically added.

3. Adjustments and Leap Years**a. Solar Calendar:**

- **Leap Year:** To account for the extra 0.24 days per year, a leap year is added approximately every 4 years (e.g., February 29 in the Gregorian calendar).
- **Consistency:** Provides a relatively stable alignment with the solar year and seasonal changes.

b. Lunar Calendar:

- **Leap Month:** An additional month is inserted approximately every 2 to 3 years to synchronize the lunar calendar with the solar year. This keeps the lunar calendar in alignment with the solar seasons.
- **Shifts:** Without these adjustments, lunar months would gradually shift relative to the solar year, causing seasonal drift.

4. Cultural and Religious Use**a. Solar Calendar:**

- **Civil Use:** Predominantly used for civil purposes, including government, business, and education. The Gregorian calendar is the most widely used solar calendar worldwide.
- **Festivals:** Solar calendars often determine fixed-date festivals and observances, such as Christmas (December 25) and New Year's Day (January 1).

b. Lunar Calendar:

- **Religious and Cultural Use:** Often used for religious and cultural purposes. For example, Islamic (Hijri) calendar is purely lunar, and Hindu and Chinese calendars are lunisolar.

- **Festivals:** Festivals in lunar calendars, like Eid al-Fitr in the Islamic calendar or the Chinese New Year, vary in date each year relative to the Gregorian calendar.
- **Islamic Calendar:** A purely lunar calendar with 12 months of 29 or 30 days each. The Islamic year is about 354 or 355 days long, resulting in annual shifts of about 10 to 12 days relative to the Gregorian calendar.
- **Time Measurement:** Solar calendars are based on the Earth's orbit around the sun, while lunar calendars are based on the moon's phases.
- **Year Length:** Solar years are about 365.24 days; lunar years are about 354 days.
- **Month Length:** Solar months have a fixed number of days, while lunar months are about 29.5 days long.
- **Adjustments:** Solar calendars use leap years to align with the solar year, while lunar calendars use leap months to keep in sync with the solar year.

Concept of a Lunisolar Year

a. Lunar and Solar Cycles:

- **Lunar Cycle:** A lunar month is based on the moon's phases, approximately 29.53 days long. Twelve lunar months total about 354 days, which is about 11 days shorter than the solar year.
- **Solar Cycle:** A solar year is based on Earth's orbit around the sun, lasting about 365.24 days. This cycle aligns with the seasons.

. Lunisolar Alignment:

- **Objective:** The goal of a lunisolar calendar is to keep the lunar months in sync with the solar year, so that the months and seasons remain aligned.

2. Structure and Adjustment

a. Months:

- **Length:** A lunisolar calendar typically has 12 lunar months, each lasting about 29 or 30 days. This results in a total of approximately 354 days per year.
- **Leap Month:** To align the lunar calendar with the solar year, an additional month (leap month) is inserted every 2 to 3 years. This ensures that the calendar remains in sync with the seasons.

b. Leap Year:

- **Purpose:** The insertion of a leap month compensates for the discrepancy between the lunar year (about 354 days) and the solar year (about 365 days). This adjustment keeps the calendar aligned with the agricultural and seasonal cycles.
- **Frequency:** Leap months are added about every 2 to 3 years, based on astronomical observations and calculations.

3. Example of Lunisolar Calendars

b. Hindu Calendar:

- **Structure:** The Hindu calendar is a lunisolar calendar used in India. It has 12 lunar months, with an extra month (Adhika Masa) added approximately every 2.5 years to keep it in sync with the solar year.
- **Festivals:** Festivals such as Diwali, Holi, and Navratri are timed according to this calendar, aligning with both lunar months and solar seasons.

b. Intercalation:

- **Definition:** The process of adding an extra month or days to correct discrepancies between the lunar and solar calendars.
- **Methods:** Different lunisolar calendars use various methods to calculate the timing of leap months, including astronomical observations and mathematical algorithms.

5. Implications and Uses

a. Cultural and Religious Practices:

- **Festivals and Rituals:** Lunisolar calendars are used to determine the dates of important cultural and religious events, ensuring that festivals and rituals occur in the appropriate season.

- **Agricultural Cycles:** Aligning the calendar with solar seasons helps in planning agricultural activities and harvests.

. Traditional Hindu Calendars

a. Vedic Calendar:

- **Historical Basis:** Derived from the Vedic texts, it was used in ancient India for agricultural, religious, and ceremonial purposes.
- **Month Names and Significance:** The months are associated with lunar constellations and include names like Chaitra, Vaisakha, Jyeshtha, and others. Each month is linked to specific rituals and festivals.
- **Lunar Phases:** The calendar tracks the phases of the moon, starting with the new moon and progressing through the waxing and waning phases.

b. Hindu Lunar Calendar:

- **Lunar and Solar Integration:** The Hindu calendar uses both lunar months (approximately 29.5 days) and solar years (about 365.24 days). The addition of an extra month (Adhika Masa) helps synchronize the lunar and solar years.
- **Seasonal Festivals:** Many Hindu festivals align with specific months, such as Diwali in Ashwin (October-November) and Holi in Phalguna (February-March).
- **Zodiac Signs:** Each month is associated with a specific zodiac sign (Rashi) like Mesha (Aries), Vrishabha (Taurus), and so on. This zodiac association affects the timing of various festivals and rituals.

c. Panchang:

- Components: A Panchang includes:
 - Tithi: The lunar day in the Hindu calendar.
 - Nakshatra: The lunar constellation or star grouping.
 - Yoga: The auspicious period formed by the combination of the positions of the sun and moon.
 - Karana: The half-day periods within a Tithi.
- Astrological Use: Panchangs are used for determining auspicious times for religious ceremonies, marriages, and other significant events.

Regional Calendars**. Marathi Calendar:**

- Regional Use: Used in Maharashtra and Goa. It also starts with Chaitra, similar to the Hindu lunar calendar.
- Festivals: Gudi Padwa is a significant New Year celebration, marked by erecting Gudis (decorated sticks) and festive processions.
- Local Variations: The Marathi calendar may have local variations in the names and observance of festivals compared to other regional calendars.

3. Indian Civil Calendar**a. Gregorian Calendar:**

- Adoption: The Gregorian calendar was officially adopted in India in 1957 for civil and administrative purposes.
- Structure: It has 12 months, with each month having 30 or 31 days, except February which has 28 days (29 in a leap year).
- Use: It is used for government operations, business, education, and international affairs.

4. Festivals and Observances**a. Lunar New Year Festivals:**

- Ugadi: Marks the beginning of the year in Andhra Pradesh, Telangana, and Karnataka. Celebrated with traditional foods like Ugadi Pachadi, which symbolizes the blend of sweet, sour, bitter, and spicy flavors.
- Gudi Padwa: Celebrated in Maharashtra with the hoisting of a decorated Gudi (flag) and traditional sweets like Puran Poli.
- Pohela Boishakh: The Bengali New Year, celebrated with processions, cultural performances, and traditional dishes like Panta Ilish (fermented rice with hilsa fish).

b. Solar Festivals:

- Makar Sankranti: Marks the end of the winter solstice and the start of longer days. Celebrated with kite flying, feasts, and bonfires.
- Pongal: A four-day harvest festival in Tamil Nadu celebrating the end of the harvest season. It includes cooking a special dish called Pongal and various cultural events.

Mathematics in Ancient India

Mathematics on the Indian subcontinent has a rich and long history going back over 5,000 years and thrived for centuries before advances were made in Europe. Its influence spread to China, Southeast Asia, the Middle East, and Europe. Apart from introducing the concept of zero, Indian mathematicians made seminal contributions to the study of geometry, arithmetic, binary mathematics, the notion of negative numbers, algebra, trigonometry, and calculus among other areas. The decimal place value system that is employed worldwide today was first developed in India. Classical period (400–1600) This period is often known as the golden age of Indian Mathematics. This period saw mathematicians such as Aryabhata, Varahamihira, Brahmagupta, Bhaskara I, Mahavira, Bhaskara II, Madhava of Sangamagrama and Nilakantha Somayaji give broader and clearer shape to many branches of mathematics.

Concept of Zero: 'Zero' was discovered by an anonymous Indian. The concept of zero in India holds significant historical and mathematical importance. The numeral zero, as we understand it today, is an integral part of the decimal numeral system, and it was developed in ancient India.

- 1. Early Development:** India's earliest recorded use of a decimal system and zero can be traced back to the Indus Valley Civilization, which existed around 2500-1500 BCE. However, the concept of zero as a number with its own value and significance started to evolve later.
- 2. Brahmagupta's Contributions:** One of the key figures in the development of the concept of zero in India was the mathematician Brahmagupta, who lived around 598-668 CE. He made significant contributions to the understanding of zero. He defined zero as a number and discussed its mathematical properties. He introduced rules for arithmetic operations involving zero, such as the addition and subtraction of zero, which are now fundamental concepts in mathematics.
- 3. Aryabhata's Work:** Aryabhata, an Indian mathematician and astronomer who lived around 476-550 CE, also played a role in zero development. In his work, Aryabhata used a symbol to represent zero and recognized its importance in mathematical calculations.

Importance in Modern Mathematics: Zero is now considered one of the most fundamental concepts in mathematics. It serves as a placeholder in the decimal system, enabling us to represent numbers of varying magnitudes efficiently. It is also crucial in algebra, calculus, and many other branches of mathematics. The concept of zero in India has a long and rich history, with mathematicians like Brahmagupta and Aryabhata making significant contributions to its development.

Concept of Pi:

The concept of π (pi), which represents the mathematical constant approximately equal to 3.14159, has a long history in India, as well as in other ancient civilizations. In India, pi was known and calculated by mathematicians and astronomers in various ways over the centuries.

1. Ancient Texts: Indian mathematicians and astronomers in ancient times made references to pi in their texts. For example, the "Shulba Sutras," which are a collection of ancient Indian texts dealing with geometry and construction, include approximations of the value of pi

2. Aryabhata: Aryabhata, an Indian mathematician and astronomer who lived around 476-550 CE, calculated an approximate value of pi as 3.1416 in his work "Aryabhatiya." He used a geometric method to estimate pi, which was later refined by other Indian mathematicians.

3. Bhaskara I: Bhaskara I, who lived in the 7th century CE, also provided an approximation of pi in his work "Mahabhaskariya." He used a different method involving a polygon with 384 sides to estimate pi more accurately.

important to note that while Indian mathematicians made notable contributions to the understanding and approximation of pi, the concept of pi and its accurate calculation were also pursued in various other ancient civilizations, such as Babylon, Egypt, and Greece. The approximate value of π is 22/7 or 3.14159265359.

Number System:

India has a rich history of number systems, with significant contributions to the development of various numeral systems. One of the most

influential contributions is the decimal numeral system, which is now the most widely used number system globally. Overview:

1. **Indus Valley Civilization:** The earliest known evidence of a numerical system in the Indian subcontinent can be traced back to the Indus Valley Civilization (around 2500-1500 BCE). Archaeological findings indicate the use of symbols for counting and recording quantities.

Decimal System:

The most significant contribution of India to the world of mathematics is the decimal numeral system. This system is based on the number ten and uses ten symbols (0-9) to represent all possible numbers. The concept of zero (0) as a placeholder and as a number itself was developed in India. The decimal system, with its place-value notation, simplifies arithmetic calculations and is the foundation of modern mathematics.

Aryabhata: In his work "Aryabhatiya," he discussed the relationships between the lengths of chords and the radii of circles, which are related to the Pythagorean theorem. While these Indian mathematicians and texts demonstrated knowledge of geometric principles related to the Pythagorean theorem, it's important to note that the theorem is named after Pythagoras because he is traditionally credited with providing the first formal proof of the theorem in ancient Greece.

Vedic Mathematics: Vedic Mathematics is a system of mathematics that claims to be based on the ancient Indian scriptures known as the Vedas. It is often attributed to a scholar named Swami Bharati Krishna Tirtha, who published a book titled "Vedic Mathematics" in 1965, where he introduced and popularized this system. However, it's important to note that the legitimacy and historical accuracy of Vedic Mathematics as a distinct mathematical system is a topic of debate among mathematicians and scholars.

Key features and claims associated with Vedic Mathematics:

1. **Sutras:** Vedic Mathematics is said to be based on a set of 16 Sutras (aphorisms) and 13 Upa-sutras (sub-sutras) extracted from the Vedas. These Sutras are claimed to provide efficient and unconventional methods for performing various mathematical operations, such as

addition, subtraction, multiplication, division, square roots, and cube roots.

2. **Mental Calculation:** Vedic Mathematics emphasizes mental calculation and claims to offer techniques that allow individuals to perform complex mathematical computations in their heads quickly. Proponents of Vedic Mathematics argue that these techniques can lead to faster and more efficient calculations compared to traditional methods.

3. **General Applicability:** Vedic Mathematics is said to be applicable to a wide range of mathematical problems and is not limited to specific areas of mathematics. Proponents claim that it can be used in various fields, including arithmetic, algebra, geometry, and calculus.

Educational Use: Vedic Mathematics has been introduced as a teaching tool in some schools and educational programs. It can help students develop mental math skills and improve their speed and accuracy in calculations. In summary, Vedic Mathematics is a system of mathematical techniques and methods that claim to be based on ancient Indian scriptures. While it has gained some popularity and is promoted as a way to improve mental calculation skills.

Famous Mathematicians in India

Aryabhata: Aryabhata was an ancient Indian mathematician and astronomer who lived in the late 5th century CE. He is one of the most significant and influential mathematicians and scientists in the history of India.

Key contributions and information about Aryabhata:

1. **Aryabhatiya:** Aryabhata is best known for his major work, the "Aryabhatiya," a Sanskrit text consisting of 121 verses. This work covers a wide range of topics in mathematics and astronomy. In the "Aryabhatiya," Aryabhata introduced innovative mathematical concepts and techniques.

2. Value of Pi (π): Aryabhata approximated the value of pi (π) as 3.1416, which was a significant improvement over earlier estimates. His work provided a more accurate understanding of the ratio of the circumference of a circle to its diameter.
3. Trigonometry: Aryabhata made significant contributions to trigonometry. He introduced trigonometric concepts and the sine function, which he referred to as "jya." His trigonometric tables and methods laid the foundation for further developments in Indian mathematics.
4. Astronomy: Aryabhata's work in astronomy included calculations related to the positions of planets, the lunar eclipse, and the solar eclipse. He proposed a heliocentric model of the solar system in which he correctly stated that the Earth revolves around the Sun. This heliocentric idea was well ahead of its time and predated similar ideas in Europe by many centuries.
5. Number System: Aryabhata made contributions to the Indian number system, including the use of a place-value notation system and the concept of zero as a placeholder. It's important to note that Aryabhata is often referred to as Aryabhata I, as there were other mathematicians and astronomers with the same or similar names who lived in later centuries. Aryabhata's contributions to mathematics, astronomy, and trigonometry remain significant, and his work is considered a cornerstone of ancient Indian science and mathematics. Aryabhata was the first of the significant mathematician-astronomers from the classical age of Indian mathematics and Indian astronomy. He treated mathematics as a different discipline.

Mahaviracharya: Mahavira, also known as Mahaviracharya or Mahavira II, was a renowned mathematician and Jain philosopher who lived in India in the 9th century CE. He made significant contributions to the field of mathematics, particularly in the area of number theory. Key contributions:

1. Ganita Sara Sangraha: Mahavira is best known for his work "Ganita Sara Sangraha," which translates to "Compendium of the Essence of Mathematics." This text is a comprehensive treatise on mathematics and is considered one of the most important works on Indian mathematics.

2. Number Theory: Much of Mahavira's work focused on number theory, including the study of prime numbers, factors, divisibility, and algebraic equations. He made important contributions to understanding the properties of numbers.

3. Prime Factorization: Mahavira developed a novel method for prime factorization, which involved a systematic procedure for finding the prime factors of a given number. His method was more efficient than earlier techniques and is still studied by mathematicians today.

4. Continued Fractions: He also made advancements in the study of continued fractions, which are expressions in which a number is represented as an infinite series of fractions. Mahavira's work on continued fractions was pioneering and contributed to the broader field of mathematics.

Bodhayana: Bodhayana, also known as Bodhayana Acharya, is an ancient Indian mathematician and scholar known for his significant contributions to mathematics. He is believed to have lived around the 6th century BCE, making him a prominent figure in the history of Indian mathematics. Bodhayana's work primarily focused on geometry and geometric principles. One of Bodhayana's most notable contributions is his commentary on the "Brahmasphutasiddhanta," which was originally written by the renowned Indian mathematician and astronomer Brahmagupta. Bodhayana's commentary provided valuable insights and explanations for the mathematical concepts and principles presented in the "Brahmasphutasiddhanta." In his commentary, Bodhayana elaborated on various mathematical topics, including arithmetic, algebra, and geometry. He expanded upon the geometric concepts and theorems found in Brahmagupta's work, which included ideas related to:

1. Geometry: Bodhayana's commentary covered various aspects of geometry, including the properties of triangles, circles, and other geometric shapes. He explained the principles of constructing altars and geometric figures.

2. Algebra: Bodhayana contributed to the understanding of algebraic concepts and equations. His commentary contained explanations of algebraic expressions and equations, demonstrating his mathematical prowess. Bodhayana's work, along with other ancient Indian mathematical texts and commentaries, played a crucial role in the development and transmission of mathematical knowledge in ancient India. His contributions, as well as those of other Indian mathematicians and scholars, helped shape the foundation of Indian mathematics and laid the groundwork for further advancements in the field.

Bhaskaracharya: Bhaskaracharya, also known as Bhaskara II or Bhaskaracharya II, was a renowned Indian mathematician and astronomer who lived in the 12th century CE. He was one of the most significant mathematicians and scholars of the medieval period in India.

Contributions:

1. Mathematics: Bhaskaracharya made important contributions to mathematics, particularly in the areas of algebra, arithmetic, and calculus. His work "Lilavati" is one of his most famous mathematical treatises. In this book, he covered various mathematical topics, including number theory, geometry, and solutions to mathematical problems.
2. Astronomy: Bhaskaracharya was also an accomplished astronomer. He wrote the astronomical treatise known as the "Siddhanta Shiromani." This work is divided into four sections: Lilavati (arithmetic and geometry), Bijaganita (algebra), Grahaganita (mathematics of the planets), and Goladhyaya (spherical trigonometry). In the Grahaganita section, he made significant contributions to understanding planetary motion and eclipses.
3. Bijaganita: Bhaskaracharya's "Bijaganita" is a comprehensive work on algebra. It covers topics like solving equations, indeterminate equations, and algebraic methods for finding square roots.
4. Calendar Reform: Bhaskaracharya proposed calendar reforms and made adjustments to the Indian calendar system. His insights into the calculation of leap years and the adjustment of months contributed to the accuracy of Indian calendars.

Srinivasa Ramanujan Ramanujan was born on 22 December 1887 into a Tamil Brahmin Iyengar family in Erode, in present day Tamil Nadu. He

was one of India's greatest mathematical geniuses. He made substantial contributions to the analytical theory of numbers and worked on elliptic functions, continued fractions, and infinite series. In January 1913 Ramanujan wrote to G H Hardy having seen a copy of his 1910 book *Orders of infinity*. In Ramanujan's letter to Hardy he introduced himself and his work. Srinivasa Ramanujan's work has found application in physics and engineering. G.H. Hardy invited Ramanujan to England. He developed rapidly as a mathematician under the guidance of G. H. Hardy, who brought him to Cambridge University in 1916. He published more than 340 mathematical papers spanning various mathematical areas.

- The film 'The Man who knew infinity' is based on Ramanujan's life.
- 2012 year was celebrated as National Mathematics Year by the Government of India.

Public Governance and Administration

Kautilya's *Arthshastra* is undoubtedly the single most revered text of seminal importance in ancient Indian political and administrative thought. Kautilya, also known as Chanakya or Vishnugupta was a great scholar of repute who played a dominant role in the overthrow of the Nanda dynasty and founding of the Mauryan empire in the 4th century B.C. He was the adviser and prime minister of Chandragupta Maurya, the dynasty's first king, whom he helped to accede to the throne in 321 B.C. in Magadha.

Arthshastra deals with the problem of "acquiring and maintaining the earth". It is thus the science which deals with the acquisition and protection of the means of livelihood. It is a treatise concerned with the political economy of the state and the structure and functions of public administration.

Arthshastra : Its nature and scope

- *Arthshastra* contains 32 paragraphical divisions and adhikaranas (sections).

- It is widely believed by scholars to be the oldest text on public administration anywhere in the world. The entire work contains 413 maxims and 6000 verses classified into 150 chapters.
- Of the 15 sections of the *Arthshastra* only four are devoted exclusively to public administration: the first, second, fifth and sixth. Though government machinery and various aspects of personnel administration were dealt with in these chapters, the focus of attention was on the duties of the King, or, what Chanakya called *Rajdharma*.
- The first section is concerned primarily with how the King must proceed if he is to choose competent and reliable ministers and set up effective internal and external security systems.
- The second *adhikarana* deals with political and economic regulatory agencies, the next two focus on family regulations, criminal justice and norms of public service. The rest of the *Arthshastra* take up other issues of state and sovereignty, international relations and war. In brief, the *Arthshastra* deals with the political, social and economic management of the state.
- Kautilya seemed primarily interested in underlying the principles of what he called *Rajdharma*: Kingship and administration in order to keep empires unified and stable. The *Arthshastra* counsels that no means is beyond the scope of the ruler to expand the territory, gain power and wealth of a nation. Kautilya may be said therefore to have subordinated ethics to politics though he did not endorse principles that ran counter to the Vedas on which the teachings of the *Dharmashastras* rested.

The *Arthshastra* views the pursuit of *artha*, as the primary goal of human existence. He indicates that *arjana* (creation) *vardhana* (increase) and *rakshana* (protection) are three important aspects of the wealth of a nation. Though Kautilya's primary emphasis is on politics or the "science of government" (*dandaniti*), the importance of this science is said to derive in considerable measure from the dependence on three other crucial sciences, viz, philosophy, theology and economics. The last plays a major role in the generation of wealth which is of crucial importance for the achievement of public welfare.



The Saptanga Theory

In his Saptanga Theory, Kautilya enumerates seven *prakritis* or essential organs of the state. They are *Swamin*, *Amatya*, *Janapad*, *Durga*, *Kosha*, *Danda* and *Mitra*.

The Swamin

Kautilya gives extensive powers to the king, along with an extensive list of duties which he must perform for the welfare of his people.

- Firstly, the king should preferably belong to the *nobility*, be a *native* of the land and follow the teachings of the *Shashtras*. Emphasis is laid on the King's training in *philosophy, economic sciences and political science*, therefore expecting every King's *legitimacy* to rest on *education and training*.
- The King's foremost duty is *rakshana* (protection) and *palana* (nurture) of his subjects. He is also asked to ensure their *Yoga-Kshema*, a broad term implying the idea of *welfare, wellbeing, prosperity and happiness*. The text asserts "In the happiness of the subjects lies the happiness of the King and in what is beneficial to the subjects, lies his own benefit"
- The king was the *head of civil, judicial and military administration*. Appointments to the most important offices were to be made by him, though a rigorous method of training was also mandatory for his civil and military personnel in order to be trained for their tasks.
- Religion was not meant to interfere in matters of state and administration.
- Among his other duties, great stress was laid on welfare administration, such as building of bridges and irrigational works, providing pastures for livestock, opening of trade routes and working of mines. The subjects were to be protected from natural calamities and anti social elements. It was the duty of the King to provide social security for the minors, the sick, the aged and the unemployed.
- Some 300 offenses, together with the fines to be imposed create the basis for an elaborate system of criminal justice administration in *Arthshastra*.
- By virtue of his power to guard the violation of the Dharmas, the king is the fountain of justice. The king who administers justice in

accordance with sacred law (*dharma*) evidence(*vyavahara*) custom (*samstha*) and edicts of king (*nyaya*) will be able to conquer the whole world.

- Kautilya was the first to emphasize the importance of secular law and recognized reasoning as a source of state law.

The Amatya

The king rules through an elaborate administrative machinery. Amatya stands for the higher echelons of administration who form an inner cabinet. The Prime Minister and the High Priest are the most important ministers. In times of emergency, Kautilya asks the king to consult the members of a wider council of ministers. The Inner Cabinet assists the king to appoint the ministers and the executive heads of departments. The important ministers in *Arthshastra* include the priest, the prime minister, the commander of the Army, the Treasurer General and the Collector General.

The Janapada

This refers to the citizens settled in the territory of a nation. The people should be industrious and followers of *dharma*. The king has to protect both land and the people from external aggression and make all possible efforts to promote the economic prosperity of the state.

The Durga

Durga is the fourth element of the state and it literally means a fortress. Kautilya has referred to only four types of durgas, they are as under: *audaka durga* (surrounded by water), *parvata durga* (made of rocks) *dhanvana* – *durga* (surrounded by desert) and *Vana – durga* (surrounded by forests).

The first two types of *durgas* were useful for protecting Janapada in case of emergency and the following two types provided shelters to the king in the event of any emergency.

It has also been said that on the frontiers of *Janapada* on all four sides the king should get such natural forts erected which may be suitable particularly from the defense point of view.

The Kosha

This is the fifth most important element. All the activities of the state depend on finance and therefore foremost attention should be given to the treasury. The state treasury should be permanent source of revenue for the state. The king is advised to take one sixth of the produce and there must be sufficient reserves of currency and valuable minerals like gold. In case of emergency, particularly in times of war, the king was at liberty to collect heavy taxes from his subjects. The chief sources of income of the state were the King's share of the land produce, customs duties and the amount collected as fines. The main tax payers were the farmers, traders and artisans.

The Danda

The King should have at his command a strong military force. The army ought to be well versed in military arts, contended and therefore loyal and patriotic. The *kshatriyas* are best for military service and the *senapati* (commander in-chief) should generally belong to this caste. Under him there are two branches, one in charge of actual defense forces controlling strategy and tactics on the field and the other in charge of supplies. A good fighting and loyal army is an invaluable asset to the King.

Mitra

The last element mentioned by Kautilya is *mitra* or allies and friends in need. A friend (inside or outside the state) equipped with all these qualities was indeed a real friend. It was indispensable for a king to make friends and enjoy the confidence of allies in order to achieve the goal of a prosperous state. Each of these seven components are organically linked. The king however remains the most important of all the *prakritis*.

The many “firsts” of Kautilya

- *Arthshastra* is the first comprehensive text on public administration in world literature
- He was the first to make Political Economy an independent discipline
- He was the first thinker to emphasize on the need for a strong political centre in India making its administration a major concern.
- First political realist in ancient Indian political thought
- First Indian political thinker to separate religion from politics giving the outline of a secular polity.
- He focused on a professional bureaucracy based on merit, rather than caste considerations. Departmental system and a detailed job classification scheme were the other innovations of Kautilya.
- Recognized the needs of a strong military administration
- *Saptanga* theory is an original contribution to political science and practice

Trade in Ancient India

The impact of trade and commerce on a nation's growth is undeniable. The expansion and progression of a country's business sector are often influenced by its geographical setting and environment. India, with the Himalayas to the north and seas to the south, has a unique advantage. The access to sea routes has facilitated the expansion of trade to different parts of the world.

In ancient times, India led the world in exporting commodities like silk, cotton, sugar, and precious stones. India's famous spices were also sought after in the West, traded via the famed spice route. These goods were traded for gold and silver from various nations. However, the advent of the Industrial Revolution in the West impacted Indian trade significantly.

Trade and commerce have been key drivers in shaping India's economy. Archaeological findings suggest that both land and sea trade significantly contributed to the growth of India's economy.

India was once the world's leading business hub, with commercial cities like Harappa and Mohenjo Daro. Ancient India had strong trade relationships with nations like Mesopotamia, trading commodities like gold, silver, terracotta pots, precious stones, pearls, etc.

Here are some factors that contributed to India's prominence in global trade and commerce:

- The Indian business community, particularly the Hindus, were known for their business acumen, resourcefulness, resilience, and integrity.
- Indian goods were renowned for their superior quality. The uniqueness of Indian products made them attractive to traders worldwide.
- The Indian community had a knack for shipbuilding, coupled with a deep understanding of sea routes, winds, and navigation. This allowed them to venture to far-off lands for trade.

Economic and Trade History of India

The economic and trade history of India is a rich and multifaceted narrative that spans thousands of years, reflecting the country's diverse culture, geographical advantages, and contributions to global commerce. Here is an overview of the economic and trade history of India:

Ancient and Medieval Periods (circa 2500 BCE - 1500 CE)

- Indus Valley Civilization (circa 2500 BCE - 1500 BCE):
- The Indus Valley Civilization, one of the world's earliest urban civilizations, engaged in trade with Mesopotamia, Egypt, and other regions.

- Trade items included pottery, textiles, metals, and agricultural products.

The Mauryan and Gupta Empires (circa 4th century BCE - 6th century CE):

- During the Mauryan and Gupta periods, India's economy prospered, with the establishment of trade routes and markets.
- The use of standardized coinage and weights facilitated trade.
- India's exports included textiles, spices, gems, and iron.

Medieval India and Maritime Trade (circa 600 CE - 1500 CE):

- India's maritime trade thrived during the medieval period, with Indian merchants sailing to Southeast Asia, the Middle East, Africa, and even China.
- The Chola Dynasty was renowned for its maritime trade connections with Southeast Asia.
- The Indian Ocean trade network connected India with the Arabian Peninsula, East Africa, and Southeast Asia, fostering the exchange of goods, cultures, and ideas.